

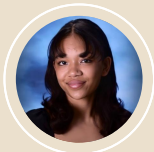
Code The Change: Article/Info Retrieval Title & Abstract Team

Team Members



Laura Vairus

Harvey Mudd College
Rising Senior
Computer Science



Alyssa Wu

Pomona College
Rising Sophomore
Computer Science + Public Policy



Wendi Zheng

Harvey Mudd College
Rising Sophomore
Mathematical and Computational
Biology

Objective

Work with the Tackle Climate Change team to aid evidence synthesization using NLP methods



Problem

- ❑ Consolidating agroforestry data
- ❑ Papers are scattered and unstandardized
- ❑ Search process is tedious and inefficient



Solution

- ❑ Natural language processing model
- ❑ Input: title & abstract of a paper
- ❑ Output: 0 or 1 for relevance

Model Testing



Training Data

Manually labeled data
from past reviews
~ 7,000 papers
~ 1,000 relevant



Text Vectorization

- Bag of words
- TF-IDF
- Word2vec
- BERT vectors
- SPECTER



Model Type

- Linear Models
- Neural Networks
- Transformers

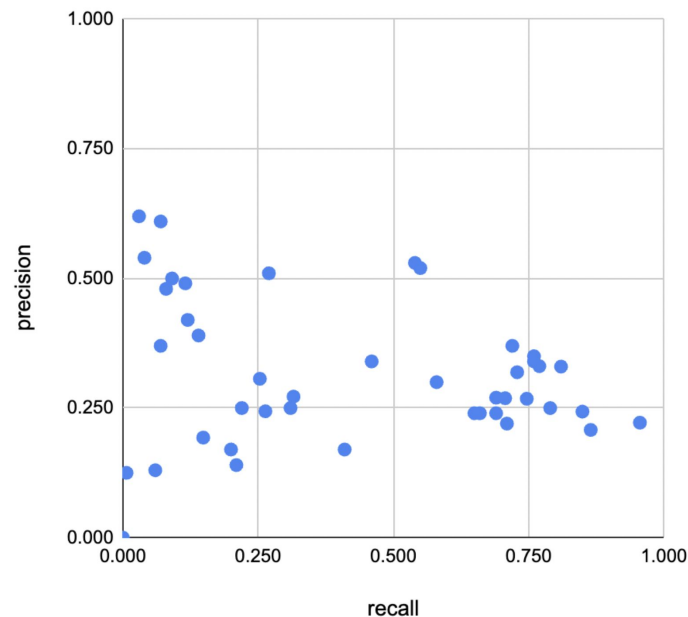


Hyperparameters

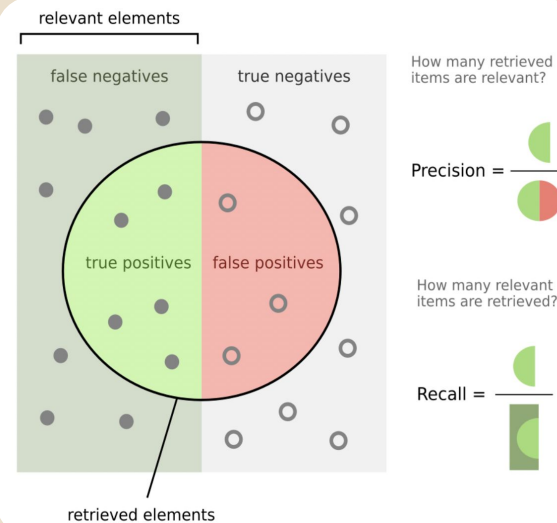
- Learning Rate
- Dropout Rate
- Batch Size
- etc.

Model Performances

Precision vs. Recall Scores of Different Models

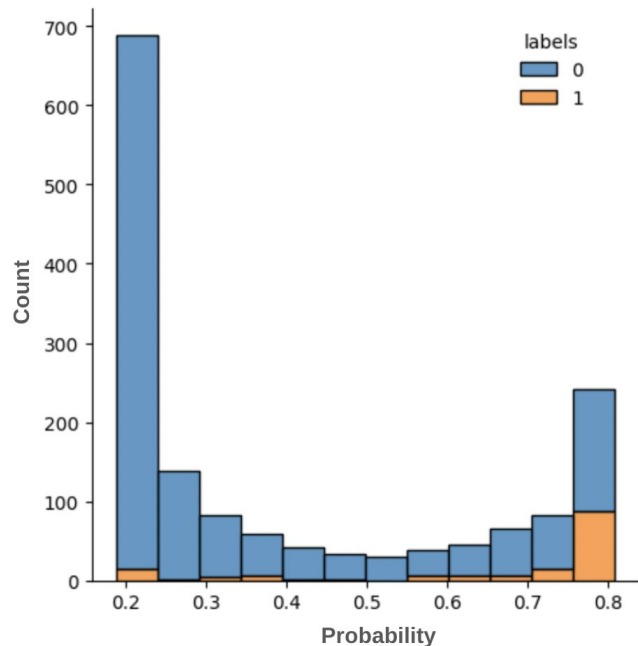


- Best recall: 0.96, best precision: 0.62
- Trade-off between precision and recall



A Closer Look: BERT-mini

Histogram of Predicted Probabilities of Relevance



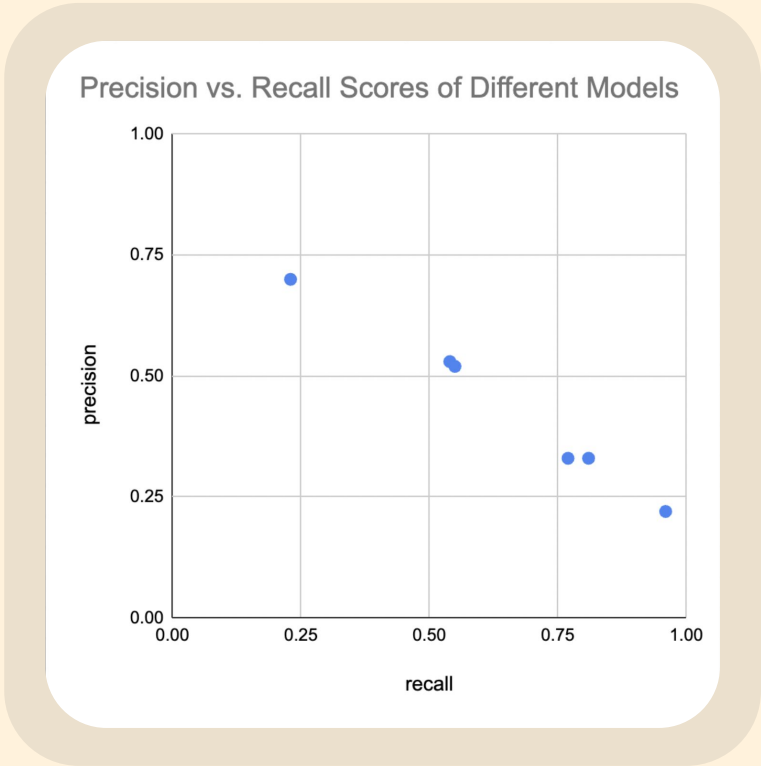
Notes

- ❑ Relevant → higher probability
- ❑ Irrelevant → lower probability
- ❑ A few relevant papers score very low and vice versa
- ❑ There are still more irrelevant papers than relevant papers in the largest bin

Key Takeaway

This model seems to recognize enough to follow the general pattern of relevance/irrelevance, but the quantity of irrelevant papers impair precision

Best Models: Pareto Optimal Front



		Precision	Recall
End-to-End Transformer Models	BERT precision	0.70	0.23
	BERT-mini	0.53	0.54
	SciBERT	0.52	0.55
Embedder + Neural Network	SPECTER NN 1 layer	0.33	0.77
	SPECTER NN 3 layers	0.22	0.96
Linear Model	Naive Bayes	0.33	0.81



Google Colab Demo

» Code The Change

- ❑ Volunteer initiative based out of Harvey Mudd College
- ❑ Provide free technological solutions to non-profits
- ❑ Seeking new projects for the coming AY25-26 year

Next Steps

Please reach out to
Code-the-change-leadership-1@g.hmc.edu
or check out
<https://codethechange.cs.hmc.edu/>



Questions?